

CC-215

Database Systems

— Unit 1 —

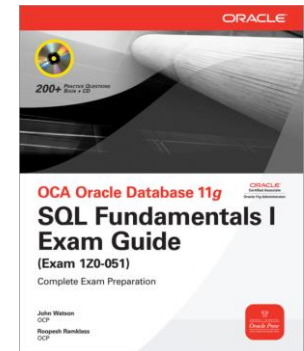
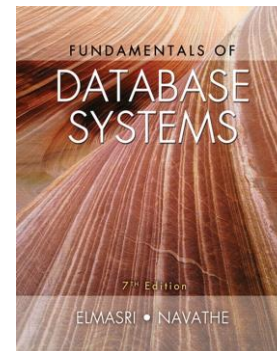
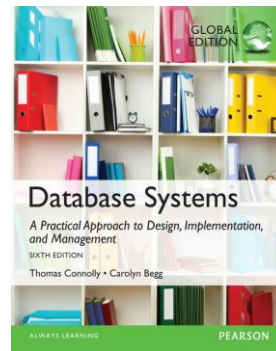
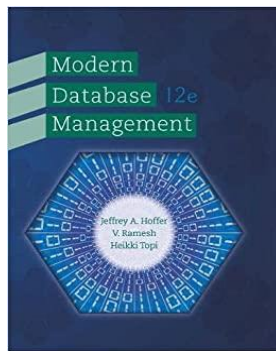
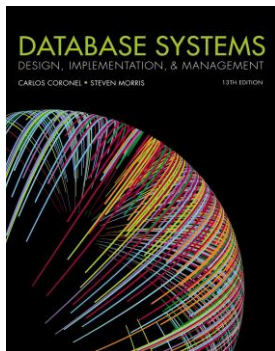
Introduction

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Department of Information Technology, FCIT

Text & Reference Books

- Database Systems – Design, Implementation and Management: Carlos Coronel, Steven Morris (13th Ed.)
- Modern Database Management: Jeffery. A. Hoffer (12th Ed.)
- Database Systems: Thomas M. Connolly (6th Ed.)
- Fundamentals of Database Systems: Elmasri (7th Ed.)
- Database System Concepts: Abraham Silberschatz (7th Ed.)
- OCA Oracle Database SQL Fundamentals Exam Guide
- Oracle Database 11g SQL: Jason Price
- Introduction to PL\SQL: Oracle Press



Course Information

- **Resources:**

\\printsrv\\teacher data\\Asif Sohail\\Database Systems

- **Office:** (Ground Floor) Graduate Building
- **Students Counseling hours :**
 - **Mon, Wed: 11:00 – 12:30 PM**
- **email:** asif@pucit.edu.pk

Assessment Category	Marks
Quizzes	12
Assignments	6
Presentations	2
Project	5

Course Overview

- Introduction
- File Processing System (FPS)
- Database Architecture and Components
- Structured Query Language (SQL)
 - Data Retrieval Language – DRL (SELECT Statement)
- Database Development and Data Modeling
- Relational Data Model and Relational Keys
- Integrity Constraints and Rules
- SQL Functions
- Entity-Relationship (E-R) Model
- SQL Joins
- Subqueries
- Creating Database Tables (Data Definition Language - DDL)

Course Overview

- Normalization
- SQL – Data Manipulation Language (DML)
- Database Objects (Views, Indexes, ...)
- Transaction Management
- Controlling User Rights & Access (Data Control Language - DCL)
- Concurrency Control
- Procedural SQL (PL/SQL)
- Stored Procedures, Functions, Triggers
- Distributed Databases
- Big Data and NoSQL
- Data Quality and Integration
- Future Trends, Revision

Lecture Outline

- Introduction
- Data? Information? Database?
- File Processing System (FPS)
- Drawbacks of FPS
- Database System or Integrated Database Environment(IDE)
- Advantages of IDE
- Types of Databases

Introduction

- A database is a collection of related **data**.
- Data refers to the raw facts & figures of an ENTITY.
- An entity can be People, Object, Event, Process, Concept etc.
- If you don't get the data right, nothing else matters.
- The selection of data to be included in a database depends upon **organizational objectives** and **information requirements**.
- The part of real world presented in a database is called **mini-world** or **universe of discourse**.
- **Data characteristics**
 - ✓ Large volume of facts
 - ✓ Difficult to interpret
 - ✓ Not suitable for decision making
- Extraction of useful information from the raw facts is very **tedious**.
Accordingly, data is **processed** to produce **information**.

Data

Class Roster

Course: MGT 500
Business Policy

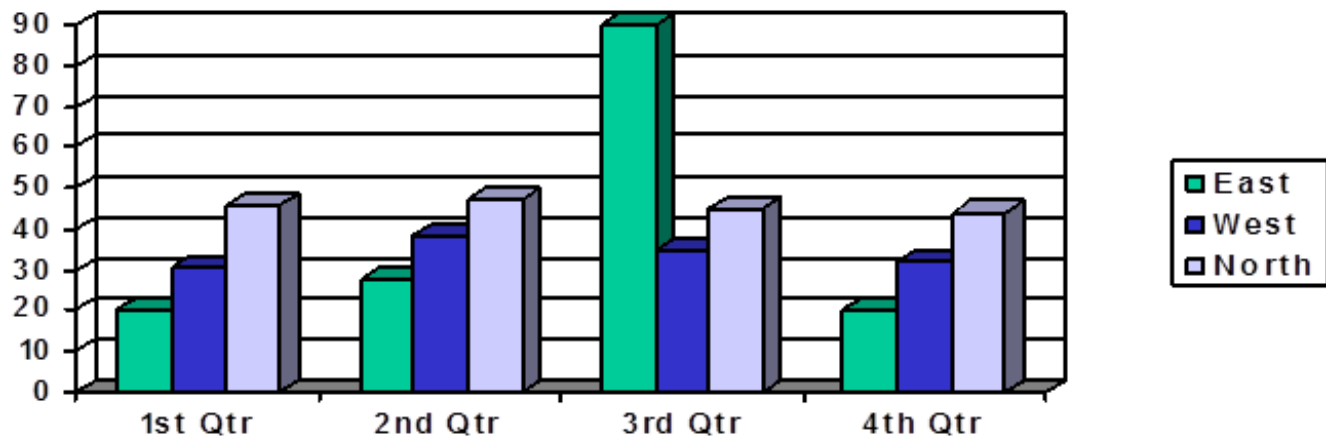
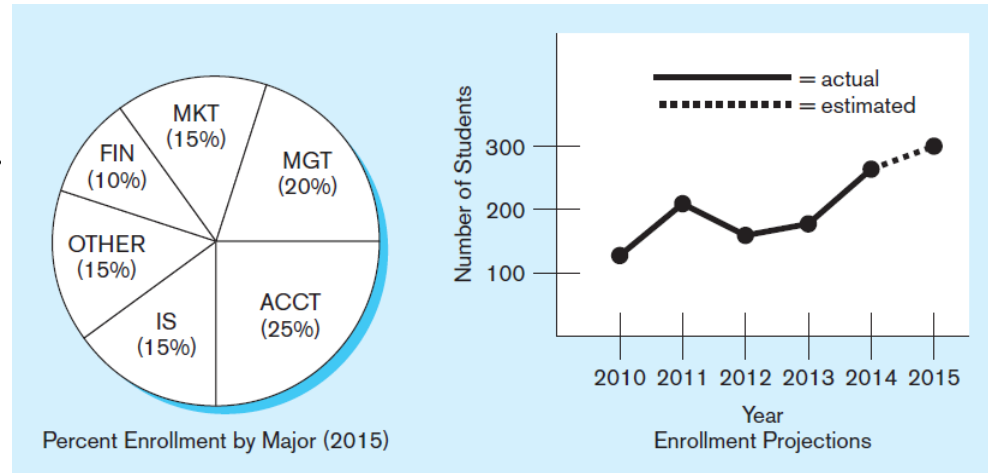
Semester: Spring 200X

Section: 2

<u>Name</u>	<u>ID</u>	<u>Major</u>	<u>GPA</u>
Baker, Kenneth D.	324917628	MGT	2.9
Doyle, Joan E.	476193248	MKT	3.4
Finkle, Clive R.	548429344	PRM	2.8
Lewis, John C.	551742186	MGT	3.7
McFerran, Debra R.	409723145	IS	2.9
Sisneros, Michael	392416582	ACCT	3.3

Data vs. Information

- Information is derived from data.
- Information is the processed data.
- The representation and amount of information depends upon the requirements of a user.
- **Drill-down, Rollup** operations.
- When data is processed and organized into a form needed for its application, it is called an **Information Product (IP)**.
- Information may be presented either Textually or Graphically or both



Information

- Information is used to reveal the meaning of data.
- Accurate, reliable, relevant and timely information is key to good decision making.
- Information is now increasingly viewed as an organizational asset (Redman, 2008) which can be used to improve organizational performance and help an organization gain a competitive advantage in the marketplace.

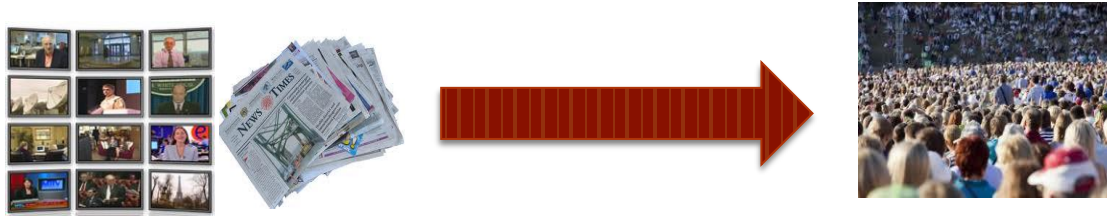
Dunn's Book of Life (1946)

- “Each person in the world creates a Book of Life. This Book starts with birth and ends with death. Its pages are made up of the records of the principal events in life. Database/Record linkage is the name given to the process of assembling the pages of this Book into a volume.”

Increasing Need of Database Systems

- The Model of Generating/Consuming Data has Changed

Old Model: Few companies are generating data, all others are consuming data



New Model: all of us are generating data, and all of us are consuming data

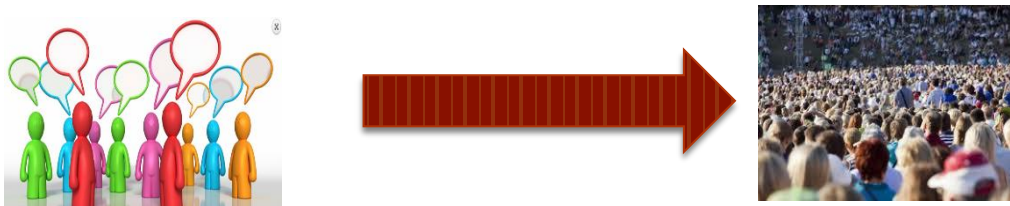


FIGURE 1.1 THE PERVASIVE NATURE OF DATABASES

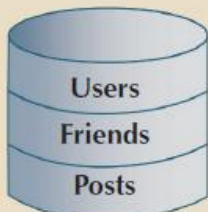
A Day In Susan's Life

See how many databases she interacts with each day

*Before leaving for work,
Susan checks her
Facebook and
Twitter accounts*



Where is the data about the friends and groups stored?
Where are the "likes" stored and what would they be used for?



*On her lunch break,
she picks up her
prescription at the
pharmacy*



Where is the pharmacy inventory data stored?

What data about each product will be in the inventory data?

What data is kept about each customer and where is it stored?



*After work, Susan
goes to the grocery
store*



Where is the product data stored?

Is the product quantity in stock updated at checkout?

Does she pay with a credit card?



*At night, she plans for a trip
and buys airline tickets and
hotel reservations online*



Where does the online travel website get the airline and hotel data from?

What customer data would be kept by the website?

Where would the customer data be stored?



*Then she makes a few
online purchases*



Where are the product and stock data stored?

Where does the system get the data to generate product "recommendations" to the customer?

Where would credit card information be stored?



Exponential Data Growth

Data in the world today

Structured data

- Databases
- Formatted files

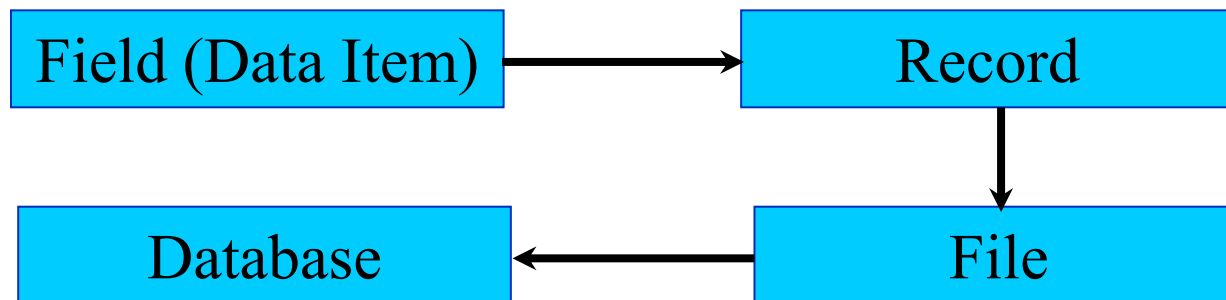
Unstructured data

- Audio
- Video
- Image data
- Documents



Data Dynamics

- The amount of available data in the modern computer world is literally exploding (3 Vs: Volume, Velocity, Variety)
- The modern data sources/types include multimedia data, sensor's data, data on web and so on.
- To get most out of data, we require tools than can simplify the task of managing the data and extract useful information out of it in timely fashion.
- **Data Hierarchy**



Database

- Database is a shared collection of logically related data, and a description of this data, designed to meet the information requirements of multiple users in an organization.
- Database is an organized collection of **data** (and **metadata**) about **entities** and **relationships** among entities.
- **Metadata:** Data that describes properties or characteristics of other data OR it is “data about data”.
- The description of data is also known as **system catalog** or **data dictionary**.
- Metadata allows database designers & users to understand what data exists and what the data mean?
- Metadata describes the **domain** (set of possible values) and the constraints for a data item.

Metadata Example

Data Item		Value			
Name	Type	Length	Min	Max	Description
Course	Alphanumeric	30			Course ID & Name
Section	Integer	1	1	9	Section Number
Semester	Alphanumeric	10	1	8	Semester & Year
Name	Alphanumeric	30			Student name
ID	Integer	9			Student ID
Major	Alphanumeric	4			Student Major
GPA	Decimal	3	0.0	4.0	Student GPA

Database System

- Database System can simply be regarded as a computerized record keeping system.
- Database System is a system whose overall purpose is to maintain data and make the information available to its users as per their requirements.
- The user of the system can perform the operations like:
 - ✓ Defining structure and types of data
 - ✓ Data manipulation (Insert, Delete, Update, Query etc.)
 - ✓ Enforcing security restrictions
- A **Database Management System (DBMS)** is a collection of programs that manages the database structure and controls access to the data stored in the database.
- A **database** and **DBMS** together are called **Database System**.

Examples of Database Applications

- Databases play a critical role in almost all areas where computers are used.
- Databases are everywhere ...
 - Student / Employee's Information System
 - Library System
 - Hotel / Airline Reservation System
 - Billing System
 - Stock / Inventory System
 - Payroll System
 - Geographical Information Systems (GISs)
 - Data Warehouses

Modes of Database Systems

- Broadly speaking, there are two modes in which databases are used.
 1. **(Online) Transaction Processing:** a large number of users use the database
 2. **Data Analytics:** processing of data to draw conclusions, and infer rules or decision procedures, which are then used to drive business decisions.
- For example, banks need to decide whether to give a loan to a loan applicant, online advertisers need to decide which advertisement to show to a particular user (Recommendation Systems). These tasks are addressed in two steps. First, data-analysis techniques attempt to automatically discover rules and patterns from data and create predictive models. These models take as input attributes (“features”) of individuals, and output predictions such as likelihood of paying back a loan, or clicking on an advertisement, which are then used to make the business decision.

Record Keeping Techniques

1. Manual Record Keeping.
2. Computerized Record Keeping
 - a) File Processing System.
 - b) Database System or Integrated Database Environment.
(The paradigm shift)

Case Study

- Consider a College System consisting of the following offices:
 - Admission Office
 - Academics Office
 - Exam Office
- Each Office is maintaining its own set of files for its day to day operations.
- Some of the files used in the system are Student's File, Faculty File, Course File, Correspondence File etc.

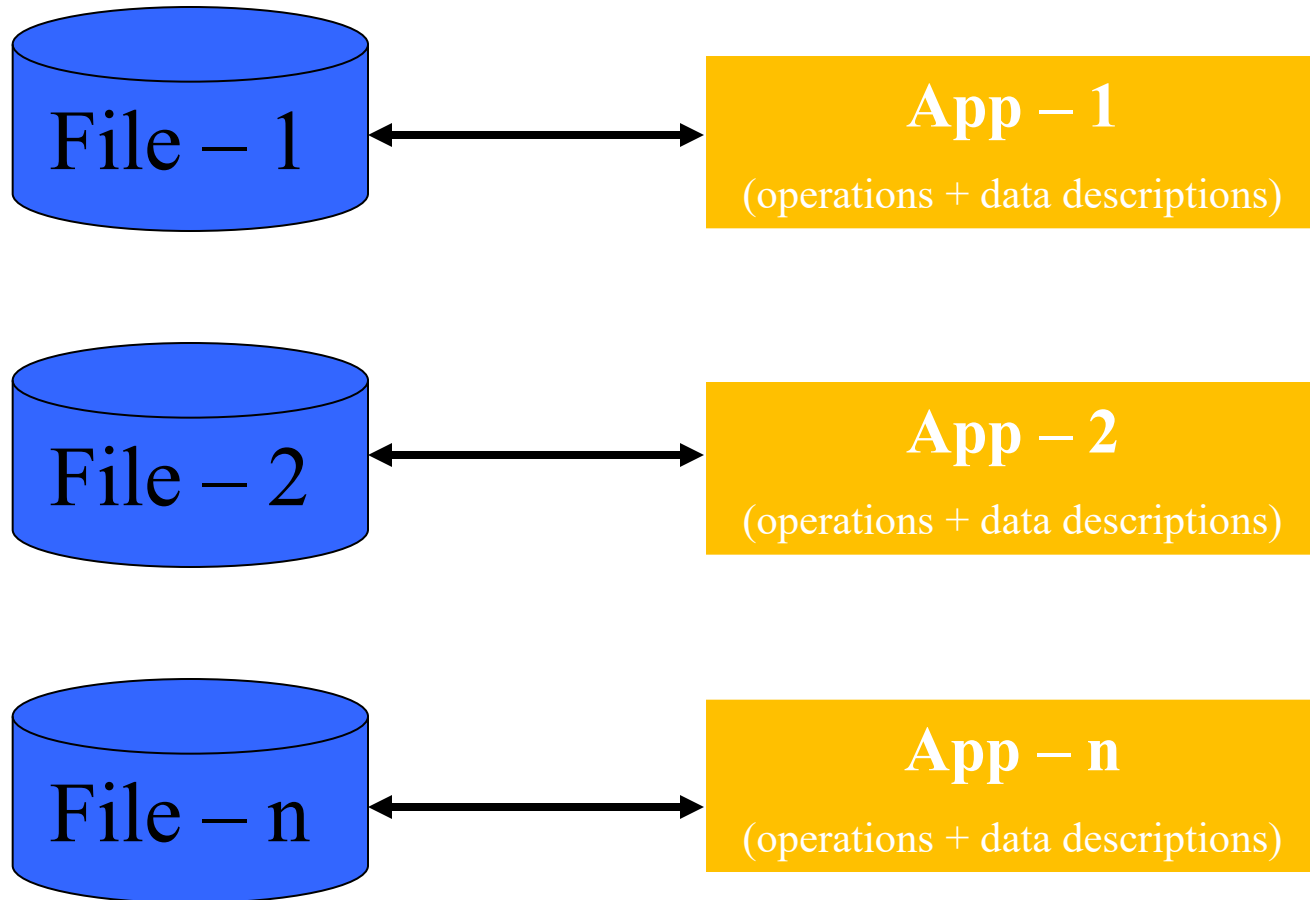
Manual System - Drawbacks

- High data volume
- Not reliable
- Inefficient
- Duplication of data
- Inconsistency
- A lot of data movement is required
- The System can't answer complex queries involving multiple departments.

File Processing System

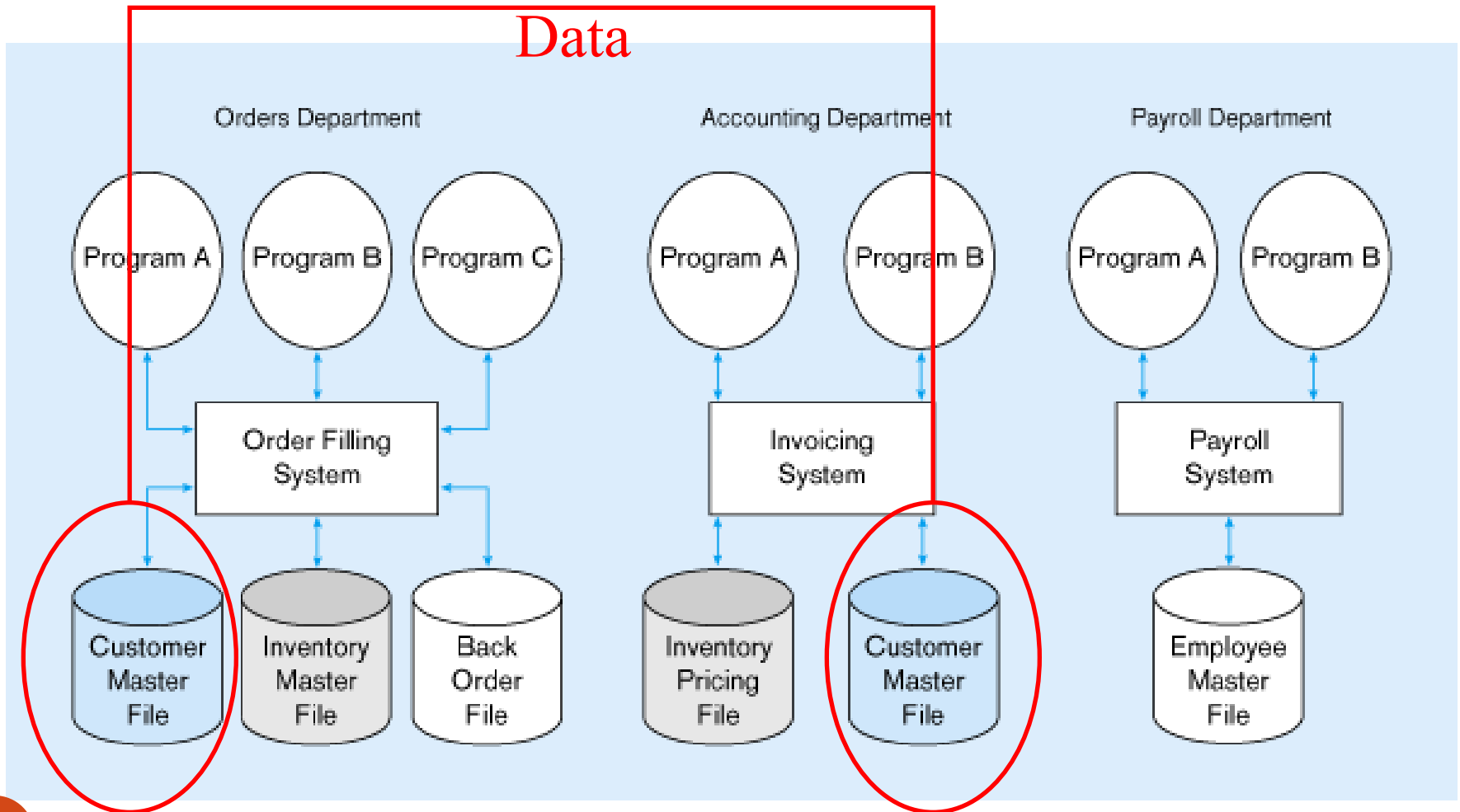
- File-based systems were an early attempt to computerize the manual filing system.
- The manual files were replaced by computer files.
- A person responsible for developing and managing computerized file processing system was called **Data Processing (DP) specialist**.
- In the traditional file processing system, the existing manual system is automated by focusing on the data processing needs of the individual departments instead of treating the organization as whole.
- Each application will have its own set of **Private Files** designed to meet the needs of a particular department.

File Processing System



File Processing System

Duplicate
Data



Drawbacks of File System

1. Redundancy of Data
2. Inconsistent Data
3. Poor Enforcement of Standards
4. Data Dependence
5. Limited Data Sharing
6. Extensive Development & Maintenance Effort
7. Limited Support for Queries
8. Inflexible
9. Lack of provision for security
10. Limited recovery from failure

Drawbacks of File System

1. Redundancy of Data

- Redundancy means duplication of data.
- Since applications in FPS are independent, so unplanned duplicate data is there.
- Same information is needed to be kept at different places. It costs time and money.
- Same data may have to be input several times to update all the occurrences.

Drawbacks of File System

2. Data Inconsistencies or Data Anomalies

- Redundancy leads to **data inconsistency** or **data anomalies**.
- A **data anomaly** occurs if an operation (update, insert, delete) has not yet been performed against all the occurrences. Consequently, same data stored at different places will disagree with each other
- The use of different programming environment and platforms lead to Heterogeneous Environment.

3. Poor Enforcement of Standards

- Organization wide enforcement of standards is poor. This leads to the following types of inconsistencies:
 - a) **Synonym**: Using different names for same data item, e.g., Stu-Id, Reg-No.
 - b) **Homonym**: Using same name for different data items, e.g., "course" for a single subject or entire program.

Drawbacks of File System

4. Structural and Data Dependence

- **Structural Dependence:** The applications are constrained to work only with the given file description. Any change in the file structure requires changes to all the applications using that file.
- **Data Dependence:** The definition of data is embedded in the application programs. Changes in the data type or length require changes in the application programs.
- Structural & Data dependence are undesirable due to:
 - Errors are often introduced when making changes.
 - It is often difficult to locate all programs affected by change
 - Incompatible file structures (use of different programming languages) makes the situation even worse.

Drawbacks of File System

5. Data Isolation or Limited Data Sharing

- As each application has its own private files so little opportunity to share data with others.
- The filing system works well when we have to interact with each file separately. However, it breaks down when we have to **cross-reference** or process the information in the files. This makes it hard to answer queries involving multiple files.

6. Extensive Development & Maintenance Effort

- The development of new application and the maintenance of the existing applications require extra efforts due to:
 - Structural and Data dependence
 - Creation of new files and/or updating the structure of the existing files.

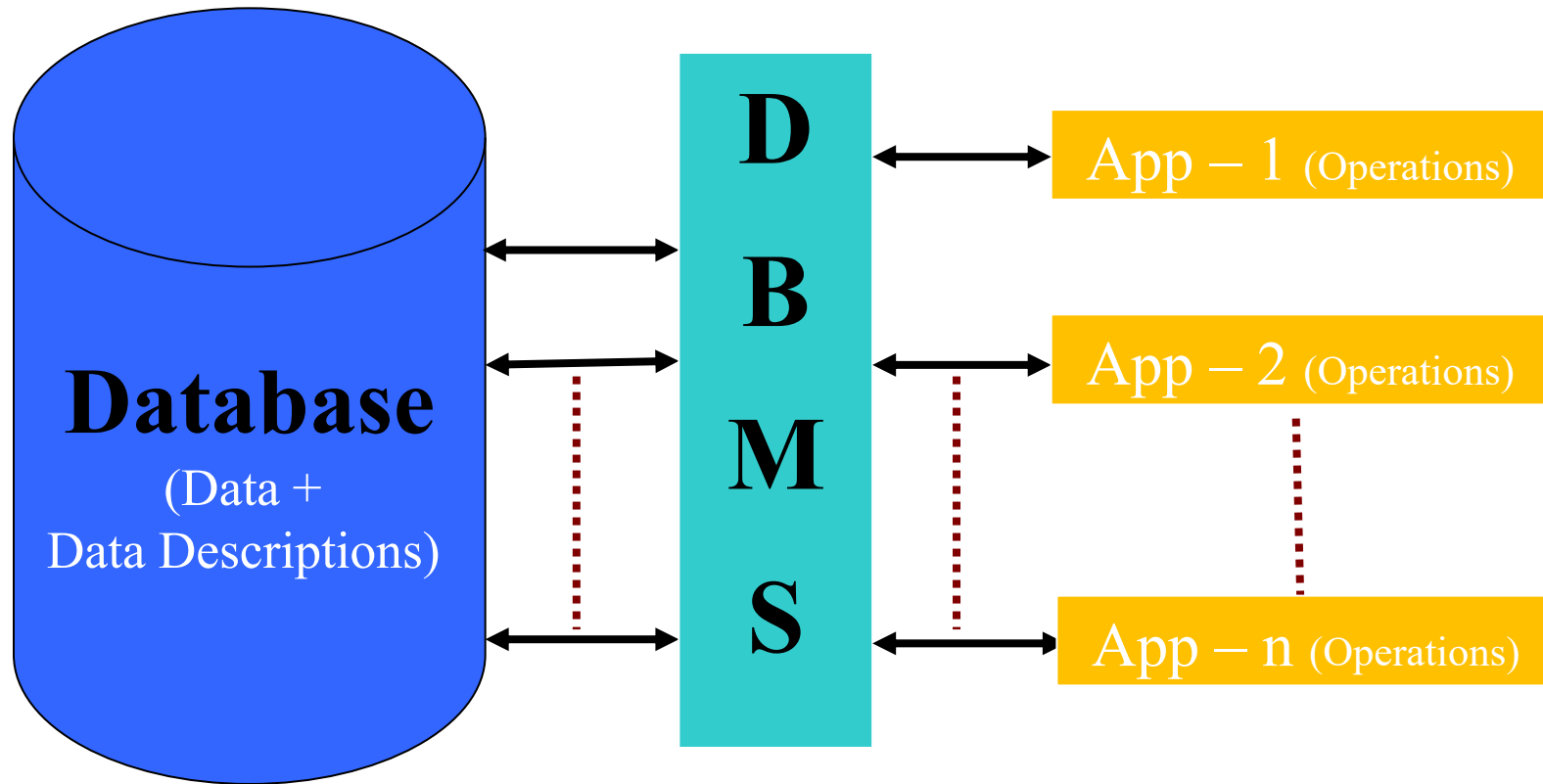
Drawbacks of File System

- 7. Limited Support for Queries
- 8. Inflexible
- 9. Lack of provision for security
- 10. Limited recovery from failure

Integrated Database Environment(IDE)

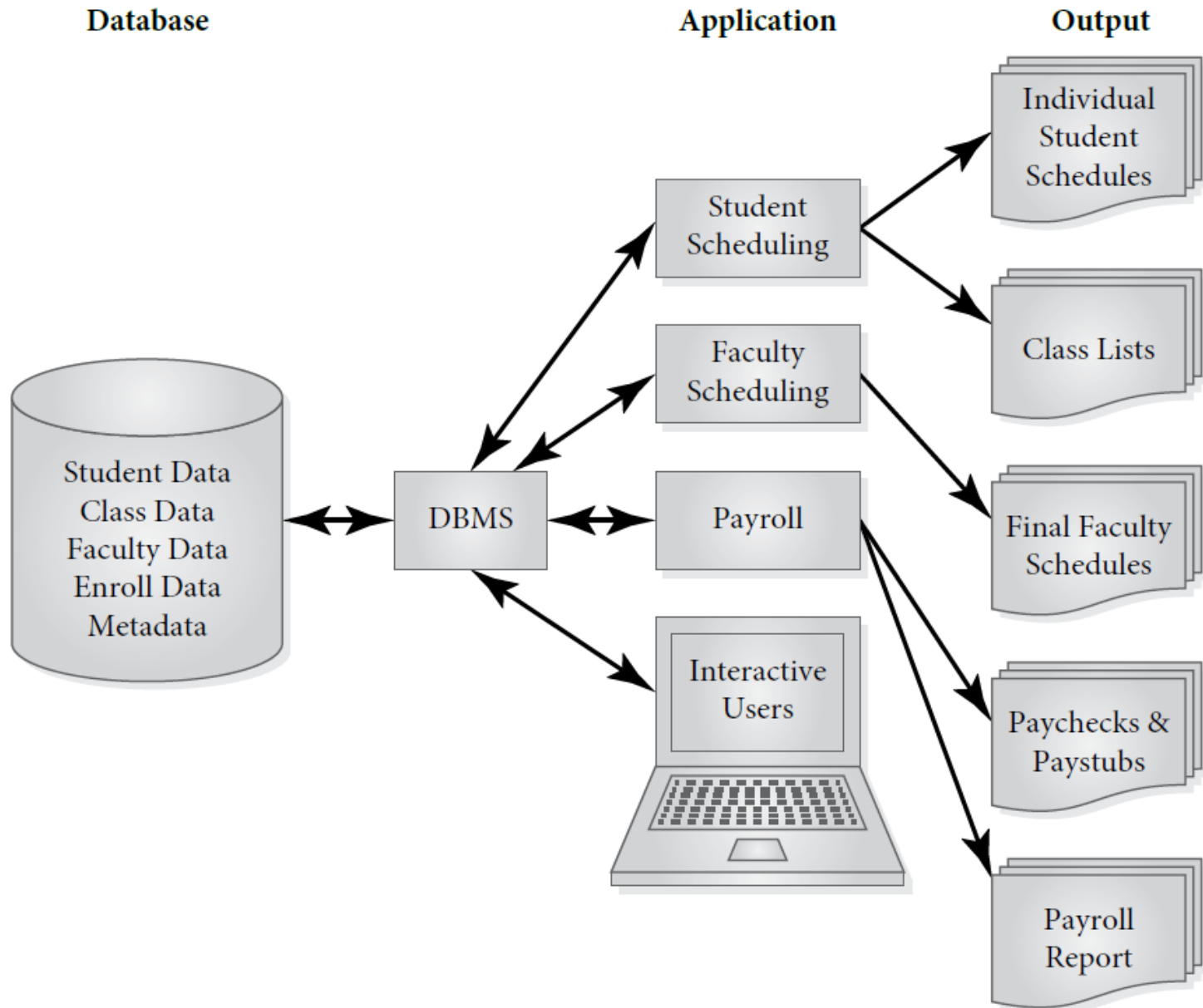
- Major factors behind the limitations of file-based approach are:
 - Redundancy of data
 - Data dependence
- In contrast to file processing system, the IDE has a single large repository of data, where data definition is separated from application programs.
- Emphasizes the integration & sharing of data throughout the organization.
- The organization wide requirements are analyzed as a whole and there is no longer concept of MY FILE or Private Files.
- A Database Management System (DBMS) is a software designed to assist in maintaining and utilizing large collections of data.
- By storing data in a DBMS rather than as a collection of files, we achieve enormous advantages.

Integrated Database Environment (IDE)



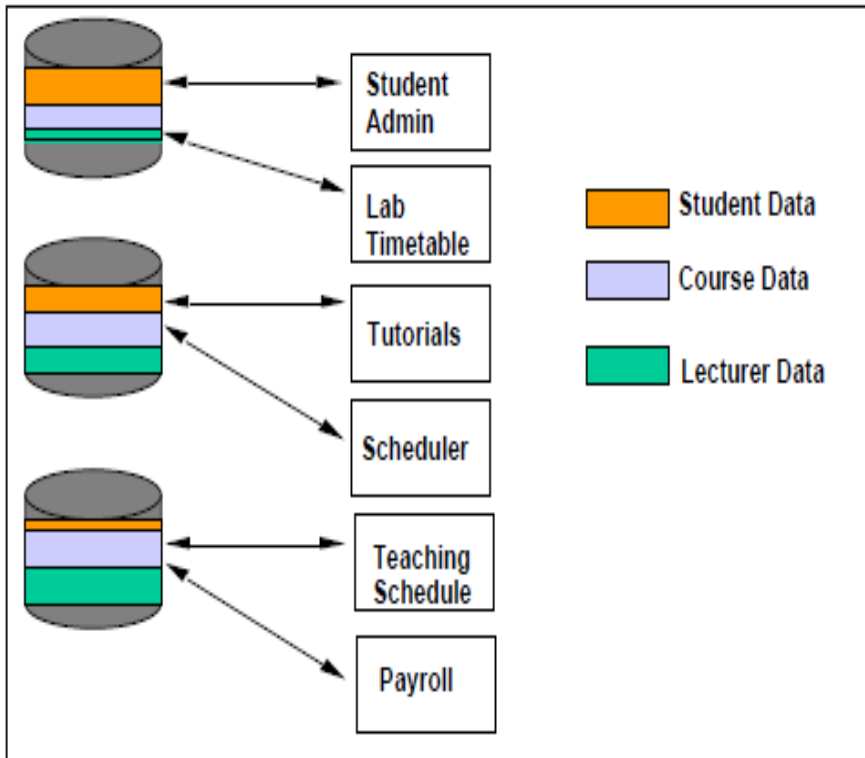
- Database system is **Self-Describing**, i.e., it contains not only the database itself but also **meta-data**: complete definition or description of the database structure and constraints.
- DBMS manages data resources like an operating system manages hardware resources

Integrated Database Environment (IDE)

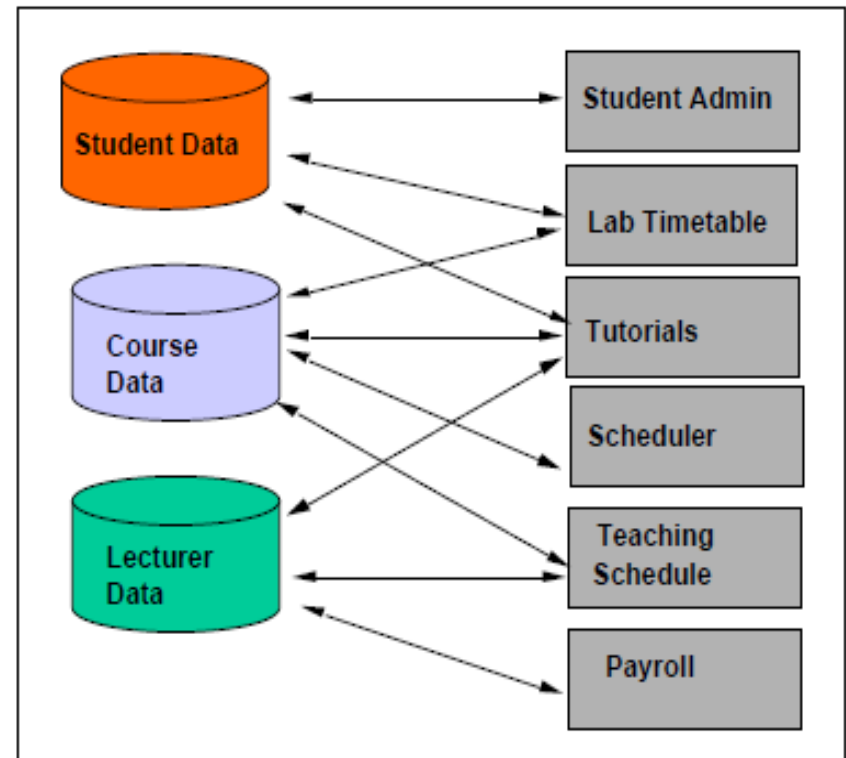


Contrasting Database and File System

File System



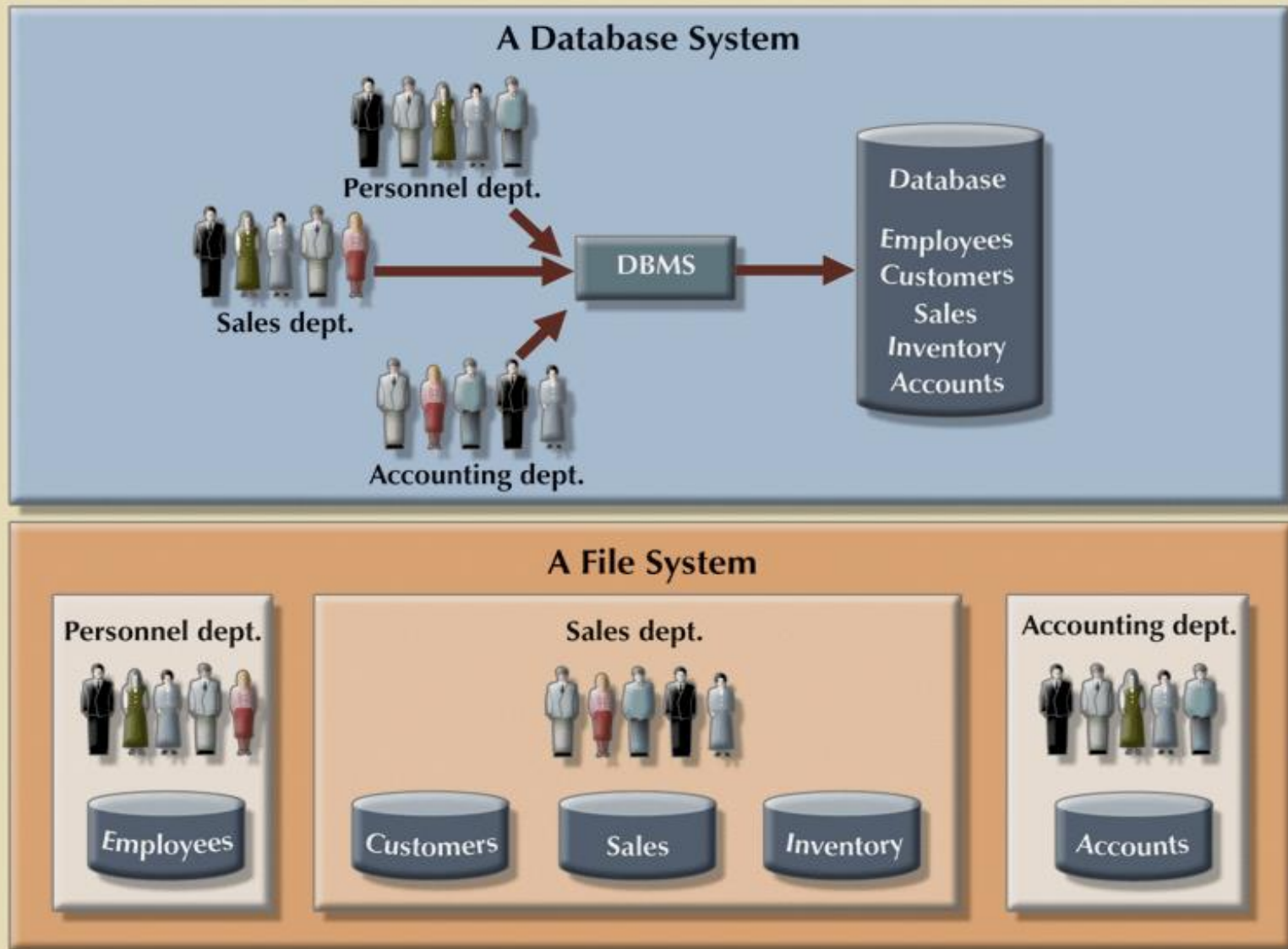
Database System



Adapted from Lecture slides of Werner Nutt

**FIGURE
1.6**

Contrasting database and file systems



Advantages of Database Approach

1. Reduced or Controlled Data Redundancy
2. Improved Data Consistency
3. Enforcement of Standards
4. Reduced Program Maintenance
5. Data Sharing -> Economy of Scale
6. Data Integrity -> Improved Data Quality
7. Improved Security Restrictions
8. Improved Accessibility & Responsiveness
9. Improved Decision Making
10. Efficient Query Processing
11. Backup & Recovery
12. Concurrency Control
13. Data Independence

Advantages of Database Approach

1. Reduced or Controlled Data Redundancy

- Since all the data items (data files) are integrated/stored in a single database, therefore redundancy is minimized.
- Limited redundancy does exist for maintaining logical connections among the related data items.

2. Improved Data Consistency

- Reduction in redundancy automatically avoids inconsistency.
- Wang and Wand describes data consistency as “a data value can only be expected to be same for the same situation”.
- If a data item appears only once, any change to its value needs to be performed only for once and the database will always be in some consistent / correct state.

Advantages of Database Approach

3. Enforcement of Standards

- This is possible as database is designed to meet the organization wide requirements.
- The standards can be name of data items and their format, data codes, documentation standards, operation standards, security policies etc.

4. Reduced Program Maintenance & Development Cost

- When all the organization's requirements are satisfied by one database instead of many separate files, the maintenance and development cost automatically reduces.

Advantages of Database Approach

5. Data Sharing

- Sharing means that the same data source is used by multiple applications.
- Data is centralized and hence can be shared not only by the existing applications but also new applications can be developed to operate against the same data. This results in additional advantages such as; faster development of new applications, and Economy of Scale.
- The applications that can access same data simultaneously are called Online Transaction Processing (OLTP) applications and DBMS uses Concurrency Control Mechanism to ensure that multiple users can access and update data correctly.

Advantages of Database Approach

6. Data Integrity (Improved Data Quality)

- Data integrity refers to the correctness of data.
- Integrity constraints or rules ensure that the data stored in the database is purified and accurate.
- Integrity constraints apply to items within a record (**intra-record constraints**) or to records that are related to one another (**inter-record constraints**), or they might be **general business constraints**.

Advantages of Database Approach

7. Improved Security Restrictions

- Database security is the protection of database from unauthorized disclosure, alteration or destruction.

DBMS provides strong security measures, such as:

- Password checks
- User defined procedures
- Defining user privileges
- Restricted view of a certain portion of database
- Audit trial system
- Data encryption

Advantages of Database Approach

8. Improved Accessibility & Responsiveness

- In File Processing System, data accessibility is quite difficult as it is procedural based. You should know the detailed procedural steps. **HOWTO DO?**
- On the other hand, accessing data is lot easier in DBMS with the help of a **non procedural language** – SQL. You only need to know the simple commands or in other words **WHATTO DO?**
- SQL allows interactive access by users.

9. Improved Decision Making

- Consolidated reports
- Ad-hoc queries
- Multiple views of data

Advantages of Database Approach

10. Efficient Query Processing

- Specialized data structures and search techniques
- Indexing
- Buffering or caching module
- Query processing and optimization

11. Backup & Recovery

- The backup and recovery subsystem of DBMS provides efficient techniques for database recovery.

12. Concurrency Control

- Simultaneous access to the shared data is possible in non-conflicting manner.

Advantages of Database Approach

13. Data Independence

- The separation of data descriptions from the applications using the data is called data independence.
- Data Independence can also be defined as: “The immunity of an application to change in storage structure – which implies that the applications don’t depend on any particular storage structure”.
- Allows change & evolution of database systems without changing the application programs
- Without data independence, it is must to understand the arrangements of data elements in order to perform an operation against them.
- Different data arrangements will need different algorithms even for the same operation.

Advantages of Database Approach

13. Data Independence.

- Clearly, data dependence is not desired because of the following reasons:
 - a) Each user should be able to access data in a customized way.
 - b) Users should not have to deal directly with the physical database storage details.
 - c) Different application may need different physical structure for the efficiency of their operations.
 - d) The DBA should have freedom to change the storage structure or access technique or both in response to changing requirements without much disturbing the existing applications.

Advantages of Database Approach

13. Data Independence

- A characteristics that allows data independence is called **data abstraction**.
- A DBMS provides users with a **conceptual representation** of data that does not include many of the details of how the data is stored or how the operations are implemented.
- A **data model** is a type of data abstraction that is used to provide this conceptual representation. The data model **hides storage and implementation details** that are not of interest to most database users.

Cost & Risk of Database Approach

1. Need for new specialized personnel
2. Cost of DBMS and additional hardware cost
3. Installation cost & complexity
4. Conversion from legacy system to modern database system
5. Organizational conflict
6. Higher impact of failure
7. Not suitable for applications with tight real-time constraints

Types of Databases

- Databases can be categorized on the basis of the following criteria:

Criteria	Types
1. Number of users	• Single User Database • Multi-user Database
2. Data location	• Centralized Database • Distributed Database
3. Data usage	• Operational (OLTP) • Analytical (OLAP)
4. Form of data	• Unstructured • Structured • Semi-structured (XML)

Types of Databases

- Personal computer databases(Single User)
- Workgroup databases(Multi-user)
- Department databases (Multi-user)
- Enterprise database (Multi-user)

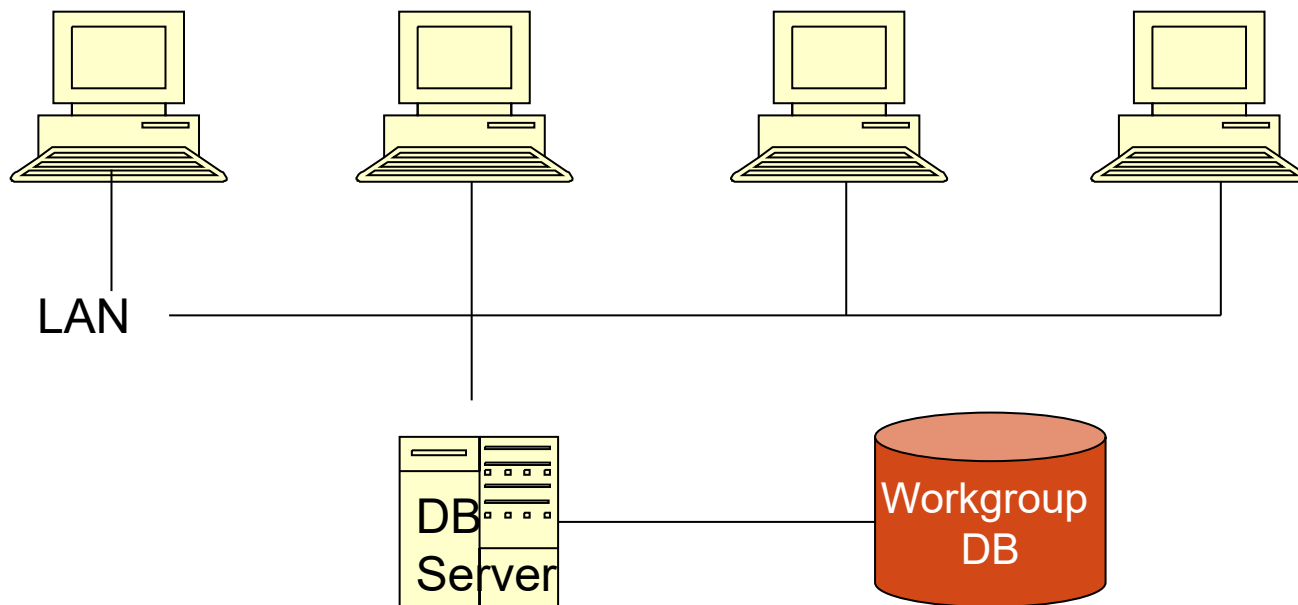
Personal Computer Databases

- Support one user with a standalone PC
- **Example:** a sales person's simple database

Types of Databases

Workgroup Databases

- Workgroup: relatively small group of people who collaborate on same project/application.
- A workgroup DB supports the collaborative efforts of a workgroup.



Types of Databases

Department Databases

- Department: Functional unit within an organization
- Department DBs support function & activities of a department.
- E.g. personnel DB to track employees, jobs, skills etc.

Enterprise Databases

- DB scope is the entire organization
- To support organization-wide operations & decision making.

Types of Databases

Internet, Intranet & Extranet Databases

- Internet: A worldwide network that connects users of multiple platforms easily through an interface known as web browser
- Extranet: Use of Internet protocols to establish limited access to company data & information by the company's customers & suppliers
- Intranet: Use of internet protocol to establish access to company data and information that is limited to the organization

Centralized & Distributed Databases

- **Centralized** database system – supports a database located at a single site
- **Distributed** database system – supports a database distributed across several different sites

Types of Databases

- The past few years have produced advances in technology leading to exciting new applications of database systems such as:
 - Multimedia databases
 - Geographic information systems
 - Data warehouses
- Multimedia databases – store pictures, video clips, and sound messages
- Geographic information systems (GIS) – store and analyze maps, weather data, and satellite images
- Data warehouses & on-line analytical processing (OLAP) is used to extract and analyze useful information from very large databases for decision-making.

Reading Material

- A. Database Systems – Design, Implementation and Management: Carlos Coronel, Steven Morris (13th Ed.)
- B. Modern Database Management: Jeffery. A. Hoffer (12th Ed.)
- C. Database Systems: Thomas M. Connolly (6th Ed.)
- D. Fundamentals of Database Systems: Elmasri (7th Ed.)
- E. Database System Concepts: Abraham Silberschatz (7th Ed.)
- F. Databases Illuminated: Catherine M. Ricardo (3rd Ed.)

Book	Section(s) / Chapter
A	1.1 to 1.3, 1.5, 1.6, 1.8
B	Chapter 1 (page 2 to 15, 27 to 30)
C	1.1 to 1.6
D	Chapter 1
E	1.1,1.2
F	1.1 to 1.3, 1.5, 2.1, 2.2



Thank you for your attention.

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